

Variables – Simple Hinglish Summary

Source: <https://javascript.info/variables>

Main baat

JavaScript application ko zyadatar **information** ke saath kaam karna padta hai. Jaise online shop me products aur cart, ya chat app me users aur messages. Is information ko store karne ke liye **variables** use hote hain.

1. Variable kya hai?

- Variable ek "**named storage**" hai, yaani data ka naam wala dabba.
- Variable banane ke liye `let` keyword use karte hain.

javascript

```
let message;           // variable declare kiya
message = 'Hello';      // value store ki (= assignment operator)
```

- Declare aur assign ek hi line me bhi kar sakte ho:

javascript

```
let message = 'Hello!';
alert(message); // Hello!
```

- **Ek line me multiple variables** bhi ban sakte hain, lekin **recommend nahi**. Readability ke liye **ek line me ek variable** likho:

javascript

```
// recommended
let user = 'John';
let age = 25;
let message = 'Hello';
```

`var` vs `let`

Purane code me `var` milta hai. Ye `let` jaisa hi hai lekin "**old-school**" tarike se. Abhi use nahi karte. Iske differences alag chapter ("The old var") me aayenge.

2. Real-life analogy

Variable ko ek **dabba (box)** samjho jis par **naam ka sticker** laga hai. Dabbe me koi bhi value rakh sakte ho aur **kitni bhi baar badal sakte ho**. Value badalne par purana data hat jata hai.

javascript

```
let message;  
message = 'Hello!';  
message = 'World!'; // value change ho gayi  
alert(message); // World!
```

Ek variable se dusre me copy:

javascript

```
let hello = 'Hello world!';  
let message;  
message = hello; // copy  
  
alert(hello); // Hello world!  
alert(message); // Hello world!
```

Do baar declare karna error hai

Variable ko **sirf ek baar** declare karo. Dobara `let` likhne par error aata hai:

javascript

```
let message = "This";  
let message = "That"; // SyntaxError: 'message' has already been declared
```

Dobara use karna ho to bina `let` ke sirf naam likho.

Functional languages (extra info)

Haskell jaisi languages me variable ki value **change nahi kar sakte**. Nayi value ke liye naya variable banana padta hai. Phir bhi ye serious kaam ke liye capable hain.

3. Variable naming rules

Do limitations:

1. Naam me sirf **letters**, **digits**, **\$** aur **_** ho sakte hain.
2. Pehla character **digit nahi** hona chahiye.

javascript

```
let userName;    // valid
let test123;     // valid
let $ = 1;       // valid
let _ = 2;       // valid

let 1a;          // galat, digit se shuru nahi ho sakta
let my-name;     // galat, hyphen (-) allowed nahi
```

- **camelCase** use karo (multiple words ke liye): `myVeryLongName`
- **Case matter karta hai:** `apple` aur `APPLE` do alag variables hain.
- **Non-Latin letters** (Hindi, Chinese, etc.) technically allowed hain, lekin **English use karna recommended** hai, kyunki code koi bhi padh sakta hai.
- **Reserved words** naam nahi ho sakte (jaise `let`, `class`, `return`, `function`):

javascript

```
let let = 5;    // error
let return = 5; // error
```

Bina `let` ke assignment (bura tarika)

Bina `use strict` ke `num = 5;` likhne se bhi variable ban jata hai (purane code ki wajah se).
Ye **bad practice** hai. **Strict mode me error** aata hai:

javascript

```
"use strict";
num = 5; // error: num is not defined
```

4. Constants (`const`)

Jis variable ki value **kabhi change nahi honi**, use `const` se declare karo. Dobara value assign karne par **error** aata hai.

javascript

```
const myBirthday = '18.04.1982';
myBirthday = '01.01.2001'; // error, reassign nahi kar sakte
```

Uppercase constants

Jo values **run hone se pehle hi pata hain** aur yaad rakhna mushkil hai (jaise color codes), unke liye **CAPITAL letters aur underscore** wale naam use karte hain.

javascript

```
const COLOR_RED = "#F00";
const COLOR_ORANGE = "#FF7F00";

let color = COLOR_ORANGE;
alert(color); // #FF7F00
```

Fayde:

- `COLOR_ORANGE` yaad rakhna `"#FF7F00"` se aasan hai.
- Spelling galti kam hoti hai.
- Code padhne me matlab clear rehta hai.

Kab CAPITAL, kab normal naam?

- **CAPITAL:** jab value **pehle se hard-coded** ho (jaise `COLOR_RED`).
- **Normal naam:** jab value **run-time par calculate** ho, chahe wo baad me change na ho.
Jaise `const pageLoadTime = ...;`

5. Naam sahi rakho (bahut important)

Variable naam ka **saaf matlab** hona chahiye. Naming programming ki sabse important skills me se ek hai. Naam dekhkar pata chal jata hai ki code beginner ne likha hai ya experienced ne.

Acche rules:

- **Human-readable** naam: `userName`, `shoppingCart`
- **Chhote abbreviations** (`a`, `b`, `c`) se bacho
- **Descriptive** naam rakho. `data` aur `value` jaise naam kuch nahi batate.
- **Team me terms fix rakho.** Visitor ko "user" kehte ho to `currentUser`, `newUser` likho (`currentVisitor` nahi).

Reuse ya naya banayein?

Kuch log naya variable banane ki jagah **purana reuse** karte hain. Isse variable ek aisa dabba ban jata hai jisme **kuch bhi** pada hota hai aur pata nahi andar kya hai. Thoda time bachta hai lekin **debugging me 10 guna zyada** lagta hai.

Extra variable achha hota hai, bura nahi. Modern browsers code optimize kar lete hain, performance ki problem nahi hoti.

Tasks (Practice)

1. Variables ke saath kaam:

javascript

```
let admin, name;
name = "John";
admin = name;
alert( admin ); // "John"
```

2. Sahi naam dena:

javascript

```
let ourPlanetName = "Earth";
let currentUserName = "John";
```

3. Uppercase const?

- `birthday` → **UPPERCASE ho sakta hai** (value hard-coded hai)
- `age` → **lowercase rahega** (run-time par calculate hoti hai, saal ke saath badalti hai)

Quick Summary

Keyword	Matlab
<code>let</code>	Modern variable, value badal sakte hain
<code>const</code>	Value change nahi ho sakti
<code>var</code>	Purana tarika, abhi use nahi karte

Topic	Yaad rakhne wali baat
Naming	Letters, digits, <code>\$</code> , <code>_</code> allowed. Digit se shuru nahi
Style	camelCase use karo
Declare	Sirf ek baar
Naam	Saaf aur descriptive rakho
Constants	Hard-coded values ke liye <code>UPPER_CASE</code>